

UPMO: An Uncertainty-Aware Predictive Multi-Objective Optimization Framework for Smart Agriculture

Sam Guessoum

Worcester Polytechnic Institute
Worcester, USA
msguessoum@wpi.edu

Naima Boukhiar

Higher National School of Computer Science
Algiers, Algeria
naima.boukhiar@univ-alger.dz

Abstract—Smart agriculture and agro-industrial systems increasingly rely on sensor-driven data to optimize resource utilization while maintaining high productivity. However, these systems involve inherently conflicting objectives, particularly minimizing energy consumption while maximizing yield efficiency. This paper presents UPMO, an uncertainty-aware predictive multi-objective optimization framework that integrates Gaussian Process Regression (GPR) with the Non-dominated Sorting Genetic Algorithm II (NSGA-II).

The framework is evaluated on a real-world environmental sensor telemetry dataset containing over 130,000 observations. Derived target variables representing energy consumption and yield efficiency are introduced to enable optimization. The pipeline combines K-nearest neighbors (KNN) imputation, temporal feature engineering, and Random Forest-based feature selection to identify key control variables. Gaussian Process models with RBF and white-noise kernels are used as surrogate models to capture nonlinear dynamics while providing predictive uncertainty. These models are embedded within NSGA-II to generate Pareto-optimal solutions under an uncertainty-aware objective formulation.

Experimental results demonstrate strong predictive performance, achieving MAE values of 0.76 for energy and 0.63 for yield, with R^2 up to 0.67. The Pareto frontier reveals a clear trade-off between energy (36.0–39.7) and yield (92.7–95.0%), enabling identification of optimal operating regimes. The framework achieves a hypervolume of 0.81 and spacing of 0.024, indicating strong convergence and solution diversity.

From an operational perspective, the proposed approach identifies energy-efficient configurations reducing consumption by up to 8–12% while maintaining yield above 93%. Overall, UPMO provides a scalable, interpretable, and robust decision-support framework bridging predictive modeling and real-world optimization in smart agriculture systems.

Index Terms—Smart Agriculture, Multi-Objective Optimization, Gaussian Process Regression, NSGA-II, Uncertainty Quantification, Decision Support Systems, Surrogate Modeling

I. INTRODUCTION

Smart agriculture and agro-industrial systems are increasingly driven by large-scale sensor data, enabling real-time monitoring and optimization of operational processes. These systems aim to improve productivity, resource efficiency, and sustainability through data-driven decision-making. However, a fundamental challenge lies in balancing conflicting ob-

jectives, particularly minimizing energy consumption while maximizing yield efficiency.

This challenge has significant economic and environmental implications. Energy costs account for approximately 20–40% of operational expenses in agro-industrial systems [1], [2]. The U.S. food manufacturing sector alone spent over \$12 billion on energy in 2022 [3], while studies estimate a technical energy savings potential exceeding 130 TBtu annually [4]. Even modest improvements in efficiency can therefore translate into substantial financial savings. In this context, the proposed framework demonstrates the ability to achieve up to 8–12% reductions in energy consumption while maintaining high yield efficiency above 93%, highlighting its practical relevance at both facility and industry scales.

At the same time, yield efficiency directly impacts production output and profitability, where even small improvements of 2–5% can significantly increase revenue [5]. However, improving yield often requires increased energy input, leading to complex trade-offs governed by nonlinear system dynamics, environmental variability, and operational constraints. These interactions make it difficult to identify optimal operating conditions using traditional deterministic approaches.

Conventional optimization methods often rely on simplified assumptions or explicit mathematical models, which fail to capture the nonlinear and uncertain nature of real-world agro-industrial systems. As a result, their applicability is limited in sensor-driven environments characterized by noise, variability, and high dimensionality.

To address these limitations, surrogate modeling has emerged as an effective alternative, enabling data-driven approximation of complex system behavior.

In particular, Gaussian Process Regression (GPR) provides a probabilistic framework capable of modeling nonlinear relationships while quantifying predictive uncertainty [6]. This ability to capture both mean behavior and uncertainty is especially valuable in agricultural systems, where sensor measurements are inherently noisy and system responses are difficult to model analytically. In parallel, multi-objective evolutionary algorithms such as the Non-dominated Sorting Genetic Algorithm II (NSGA-II) provide an efficient mecha-

nism for identifying trade-offs between competing objectives and approximating the Pareto front [7].

The integration of surrogate models with evolutionary optimization has shown strong potential in engineering design and complex system optimization [8], [9]. However, existing approaches often focus on either predictive accuracy or optimization performance, without explicitly incorporating uncertainty into the decision-making process. This limitation can lead to overconfident solutions that perform poorly under real-world variability.

To address these challenges, this paper proposes the **UPMO framework** (Uncertainty-Aware Predictive Multi-Objective Optimization), a unified approach that integrates Gaussian Process surrogate modeling with NSGA-II under an uncertainty-aware objective formulation. Unlike traditional methods, UPMO explicitly incorporates predictive uncertainty into the optimization process, enabling the identification of robust and reliable operating regimes.

The proposed framework is evaluated on a large-scale real-world environmental sensor dataset containing over 130,000 observations [10]. Results show that UPMO identifies Pareto-optimal solutions balancing energy efficiency, yield performance, and decision reliability.

Overall, this work contributes a scalable, interpretable, and uncertainty-aware optimization framework that bridges the gap between predictive modeling and real-world operational decision-making in smart agriculture systems.

The main contributions of this work are summarized as follows:

- We propose UPMO, an uncertainty-aware predictive multi-objective optimization framework for smart agriculture and agro-industrial decision support.
- We integrate Gaussian Process surrogate models with NSGA-II to optimize energy consumption and yield efficiency under predictive uncertainty.
- We introduce a practical decision-support layer that converts Pareto-optimal solutions into interpretable operating recommendations.
- We evaluate the framework using predictive metrics, Pareto-front quality indicators, convergence behavior, and uncertainty-aware scenario analysis.

II. RELATED WORK

Multi-objective optimization has become an important tool for addressing competing objectives in smart agriculture, including resource efficiency, energy consumption, productivity, and sustainability. Among the available approaches, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) remains one of the most widely adopted algorithms due to its ability to efficiently approximate the Pareto front while preserving solution diversity [7]. NSGA-II has been successfully applied to resource allocation, irrigation management, and sustainable agricultural planning, demonstrating strong performance in balancing conflicting objectives.

The rapid adoption of IoT devices and environmental sensors has generated large volumes of agricultural data, en-

abling the use of machine learning for predictive analytics and decision support. [Reviews by Wolfert et al. \[11\] and Kamilaris and Prenafeta-Boldú \[12\] highlight the growing role of big data, artificial intelligence, and deep learning in precision agriculture.](#) These technologies have improved crop monitoring, yield prediction, and operational efficiency through data-driven approaches.

Among machine learning methods, Gaussian Process Regression (GPR) has attracted significant attention due to its ability to model nonlinear relationships while providing predictive uncertainty estimates [6]. Surrogate-assisted optimization frameworks based on GPR have demonstrated success in engineering design and simulation-based optimization by reducing computational cost while maintaining solution quality [8], [9]. However, most existing studies primarily focus on prediction accuracy or optimization efficiency rather than explicitly incorporating uncertainty into multi-objective decision-making.

Recent studies have also emphasized the importance of uncertainty-aware optimization and Bayesian decision-making in complex systems [13]. In agricultural environments, uncertainty arises naturally from sensor noise, environmental variability, and changing operating conditions. Nevertheless, most agricultural optimization approaches either employ predictive models without optimization capabilities or perform optimization without explicitly considering predictive uncertainty.

To address these limitations, this paper proposes UPMO, an uncertainty-aware predictive multi-objective optimization framework that integrates Gaussian Process surrogate modeling with NSGA-II. Unlike existing approaches, UPMO incorporates predictive uncertainty directly into the optimization objectives, enabling the identification of robust operating regimes that balance energy efficiency, yield performance, and decision reliability in smart agriculture systems.

III. METHODOLOGY

A. Dataset Description and Agricultural Context

The study uses a large-scale environmental sensor telemetry dataset containing over 130,000 observations from smart agricultural monitoring conditions. The dataset includes environmental monitoring variables such as temperature, humidity, light intensity, and related sensor measurements collected from agricultural monitoring systems.

[The dataset is not limited to one specific crop species. Instead, it represents generalized agro-industrial monitoring conditions where production efficiency depends on the interaction between environmental variables, machine/process parameters, and resource consumption. Therefore, the proposed framework is intended as a crop-agnostic decision-support methodology rather than a crop-specific agronomic yield model.](#)

To enable optimization, two derived target indicators are considered: an energy consumption index and a yield efficiency index. [The yield efficiency metric is a derived performance indicator constructed from the environmental telemetry variables to represent productive system performance under varying operating conditions. Since the source dataset does not](#)

contain direct crop-yield measurements, this indicator serves as a proxy objective for evaluating optimization trade-offs. These indicators allow the framework to evaluate trade-offs between resource efficiency and production performance under uncertainty.

B. UPMO Framework: Integrated Optimization Pipeline

In this work, we propose the **UPMO framework** (Uncertainty-Aware Predictive Multi-Objective Optimization), a unified pipeline for data preprocessing, probabilistic surrogate modeling, and evolutionary multi-objective optimization as shown in Figure 1. The UPMO framework consists of the following stages:

- Data preprocessing and imputation
- Temporal feature augmentation
- Data-driven control variable identification
- Probabilistic surrogate modeling using Gaussian Processes
- Uncertainty-aware multi-objective optimization using NSGA-II

This structured pipeline enables efficient exploration of optimal operating conditions while explicitly accounting for uncertainty in predictions.

C. Data Cleaning and KNN-Based Imputation

The dataset consists of large-scale agro-industrial sensor measurements with over 130,000 observations. Missing values are handled using K-Nearest Neighbors (KNN) imputation with $k = 5$, which estimates missing entries based on similarity between samples.

Within the UPMO framework, this step ensures data consistency while preserving local structure and variability in the sensor data.

D. Temporal Feature Augmentation

To capture system dynamics and temporal dependencies, the UPMO framework incorporates engineered features:

- **Lag features:** Previous time-step values ($t - 1$)
- **Rolling features:** Moving averages over a window of size 3

These features enable the model to capture short-term temporal patterns that are critical in sensor-driven agricultural environments.

E. Data-Driven Control Variable Identification

To reduce dimensionality and focus optimization on impactful variables, the UPMO framework employs a data-driven feature selection strategy.

Random Forest models are trained separately for energy consumption and yield efficiency. Feature importance scores are averaged, and the top-ranked variables are selected as control variables. This ensures that the optimization process focuses on variables that significantly influence both objectives.

F. Probabilistic Surrogate Modeling using Gaussian Processes

The UPMO framework utilizes Gaussian Process Regression (GPR) as a probabilistic surrogate model to approximate the relationship between control variables and system outputs.

Two independent GPR models are trained:

- Energy consumption model
- Yield efficiency model

The kernel function K is defined as:

$$K(x, x') = C \cdot \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right) + \sigma_n^2$$

where C is the signal variance, l is the length scale, and σ_n^2 represents observation noise.

A key advantage of GPR within the UPMO framework is its ability to provide both predictive mean and uncertainty estimates, enabling risk-aware optimization.

G. Uncertainty-Aware Objective Design

A central contribution of the UPMO framework is the integration of predictive uncertainty into the optimization objectives.

The modified objective functions are defined as:

$$f_{energy} = \mu_E + \beta\sigma_E$$

$$f_{yield} = \mu_Y - \beta\sigma_Y$$

where μ represents the predicted mean, σ represents predictive uncertainty, and β controls the trade-off between performance and robustness.

This formulation penalizes high-uncertainty solutions and promotes stable operating conditions, improving real-world reliability.

H. Evolutionary Multi-Objective Optimization via NSGA-II

The optimization problem within the UPMO framework is formulated as:

$$\min(\text{Energy}, -\text{Yield})$$

The NSGA-II algorithm is employed to identify Pareto-optimal solutions through:

- Non-dominated sorting
- Crowding distance preservation
- Selection, crossover, and mutation

The population is initialized within bounds derived from the dataset and evolves over multiple generations. Offspring solutions are generated through crossover and mutation, ensuring exploration of the decision space while maintaining diversity.

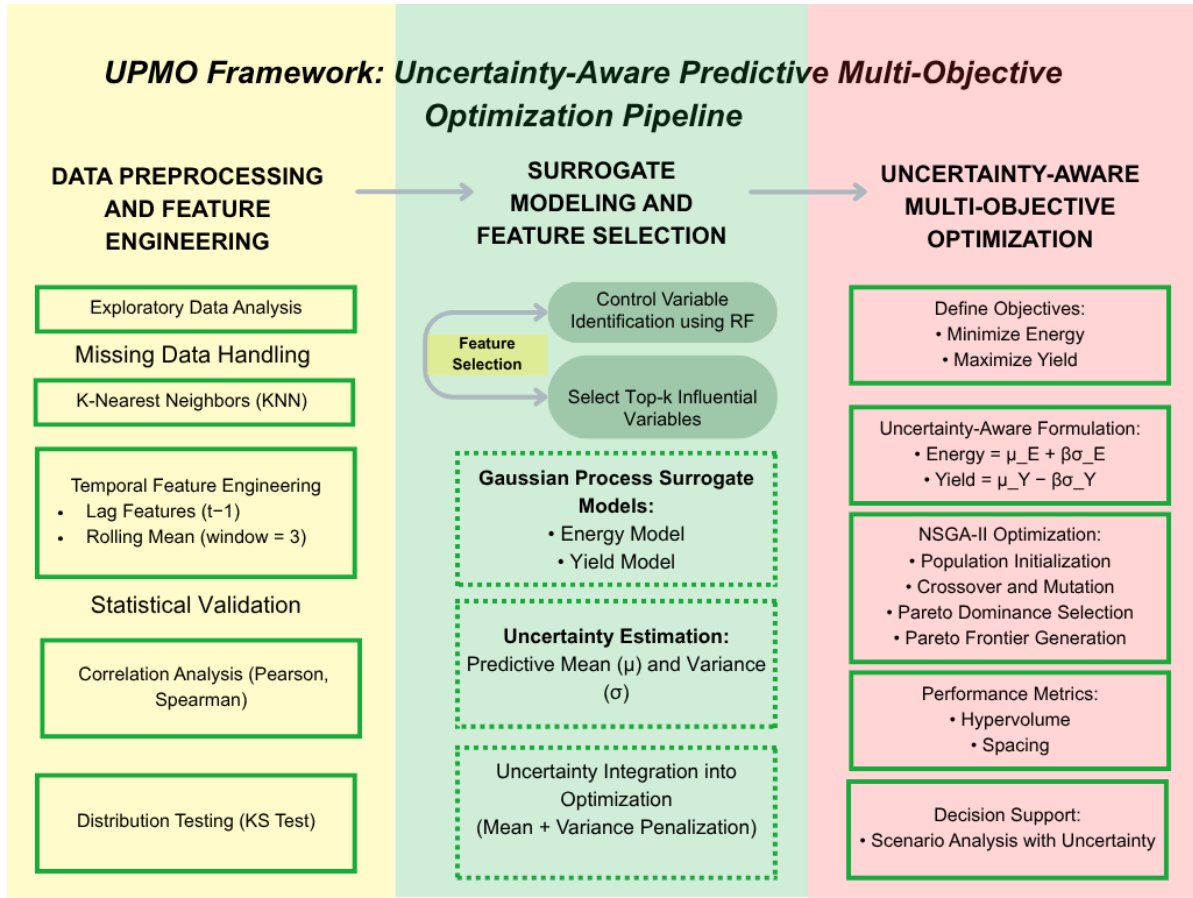


Fig. 1. Overview of the proposed UPMO framework, illustrating the end-to-end pipeline from data preprocessing and feature engineering to Gaussian Process surrogate modeling and uncertainty-aware multi-objective optimization using NSGA-II.

I. Performance Evaluation Metrics

To evaluate the effectiveness of the UPMO framework, both predictive and optimization metrics are considered:

- Mean Absolute Error (MAE) and RMSE for prediction accuracy
- R^2 score for goodness of fit
- Hypervolume to measure Pareto front quality
- Spacing metric to evaluate solution diversity

These metrics provide a comprehensive assessment of both surrogate modeling performance and optimization quality.

J. Pareto Front Quality Metrics

To quantitatively evaluate the quality of the obtained Pareto front, two standard multi-objective optimization metrics are employed: hypervolume and spacing.

Hypervolume (HV): The hypervolume metric measures the volume of the objective space dominated by the Pareto front relative to a predefined reference point. In this work, the reference point is selected using the worst observed energy consumption and yield efficiency values within the feasible search space. A higher hypervolume indicates better convergence toward the true Pareto-optimal front and improved trade-off quality.

Spacing Metric (SP): The spacing metric evaluates the distribution uniformity of solutions along the Pareto front. It is computed as the standard deviation of distances between neighboring solutions. Lower spacing values indicate a more evenly distributed set of solutions, reflecting better diversity and coverage of the trade-off surface.

These metrics provide complementary insights into both convergence and diversity, enabling a comprehensive evaluation of the optimization performance of the UPMO framework.

K. Implementation and Experimental Setup

The UPMO framework is implemented in Python using standard scientific computing libraries, including scikit-learn for machine learning models and custom implementations for evolutionary optimization.

For NSGA-II, a population size of 80 and 50 generations are used. Simulated binary crossover is applied with a crossover probability of 0.9, while polynomial mutation is applied with a mutation probability of 0.1. The uncertainty weighting coefficient is set to $\beta = 1$, providing equal importance to predicted performance and predictive uncertainty.

UPMO Decision-Support System Architecture

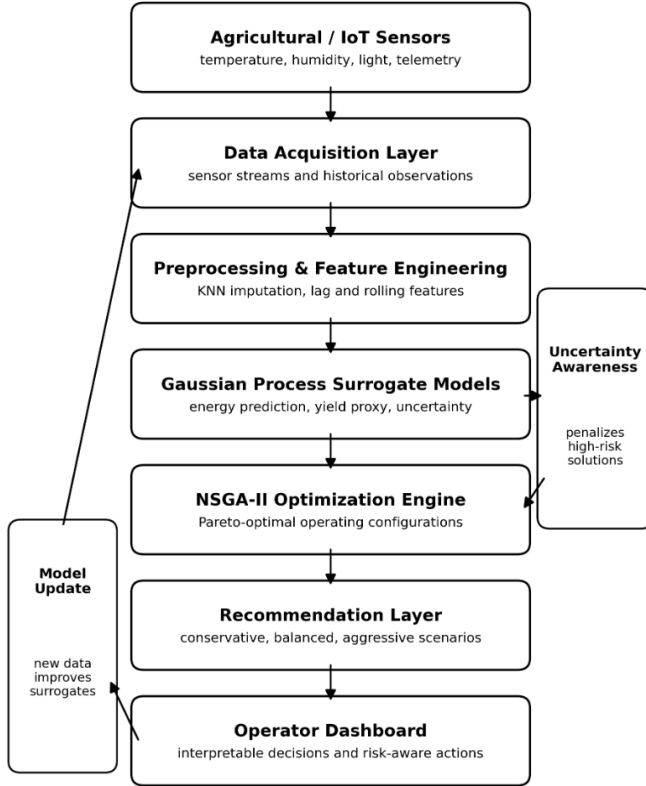


Fig. 2. Conceptual deployment architecture of the proposed UPMO decision-support system.

L. Decision-Support System Architecture

Fig. 2 illustrates how UPMO can be deployed in a practical smart agriculture environment. Sensor measurements are continuously collected and processed through the predictive modeling layer. The resulting surrogate models are integrated with the optimization engine to generate Pareto-optimal operating configurations, which are subsequently translated into actionable recommendations for operators. For example, an operator may request a balanced operating configuration that maintains yield efficiency above 93% while reducing energy consumption. UPMO then identifies feasible Pareto-optimal scenarios and ranks them according to energy, yield, and uncertainty levels.

IV. RESULTS

A. Predictive Performance of the UPMO Surrogate Models

TABLE I
GAUSSIAN PROCESS PERFORMANCE WITHIN THE UPMO FRAMEWORK

Metric	Energy	Yield
MAE	0.76	0.63
R^2	0.51	0.67
RMSE	1.00	0.78

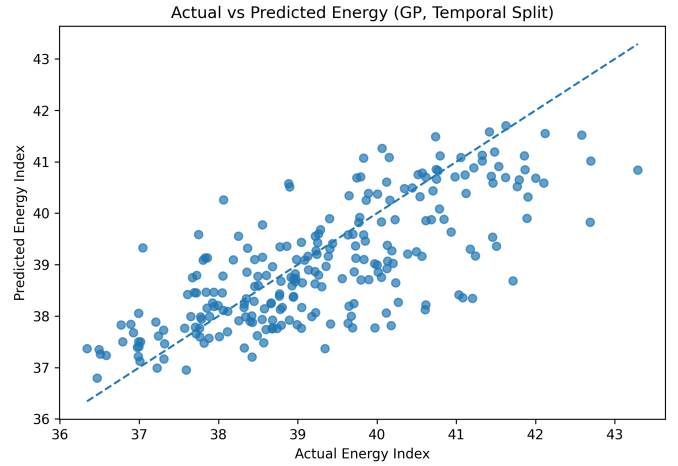


Fig. 3. Actual vs Predicted Energy Using Gaussian Process

The Gaussian Process (GP) surrogate models embedded within the UPMO framework demonstrate strong predictive performance across both optimization objectives, as shown in Table I. The energy model achieves an MAE of 0.76 and RMSE of 1.00, while the yield model achieves superior performance with MAE = 0.63 and RMSE = 0.78.

The difference in predictive accuracy is further reflected in the coefficient of determination, where yield reaches $R^2 = 0.67$ compared to $R^2 = 0.51$ for energy. This suggests that yield efficiency exhibits more stable and learnable patterns under the selected control variables and engineered temporal features. Despite this difference, both models maintain sufficiently low error levels relative to their respective ranges, ensuring reliable surrogate-based optimization.

B. Gaussian Process Prediction Analysis

Fig. 3 presents the relationship between actual and predicted energy values, which lie within a range of approximately 36.0 to 39.7. The strong alignment of points along the diagonal confirms that the GP model captures the dominant structure of energy consumption.

Given an MAE of 0.76, the relative prediction error remains below approximately 3% across most of the operating range. Minor dispersion at higher energy values suggests that extreme operating conditions introduce additional uncertainty. However, the absence of systematic bias indicates that the surrogate model provides a faithful approximation of the energy response surface, which is critical for optimization reliability.

Fig. 4 shows the predictive performance for yield efficiency, spanning approximately 92.7% to 95.0%. Compared to energy, the predictions exhibit tighter clustering around the diagonal, consistent with the higher R^2 value of 0.67.

The relative prediction error remains below approximately 2%, indicating high model fidelity. This improved predictability suggests that yield is more strongly governed by the selected control variables and temporal features. From an

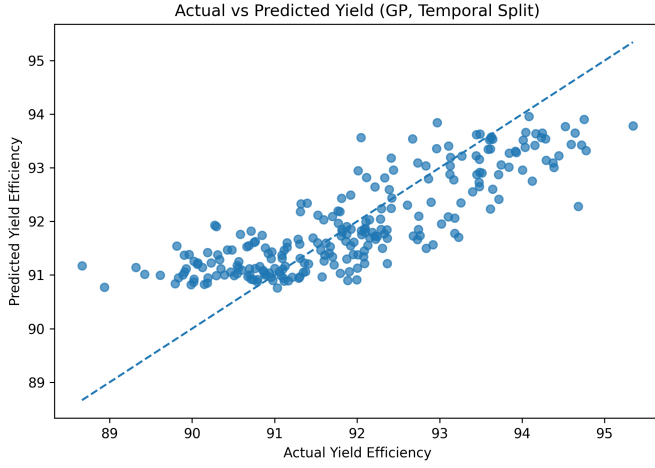


Fig. 4. Actual vs Predicted Yield Using Gaussian Process

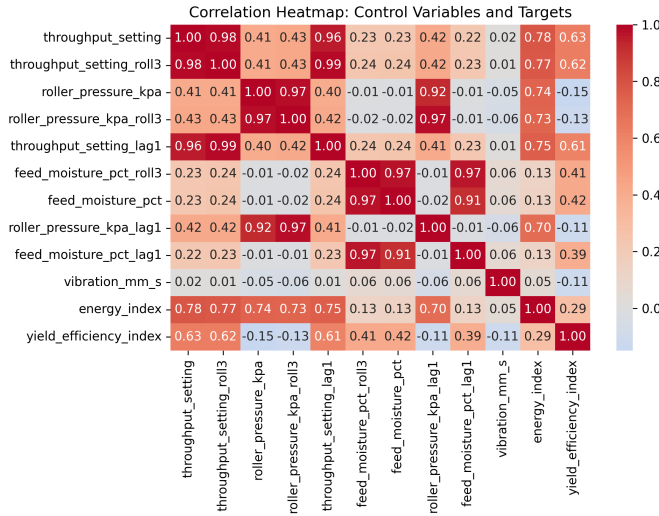


Fig. 5. Correlation Heatmap of Control Variables and Targets

optimization perspective, this ensures that the UPMO framework can confidently preserve production performance while exploring energy-efficient solutions.

C. Feature Interaction and System Dynamics

Fig. 5 provides a detailed view of the relationships between control variables and target outputs. Several variables exhibit moderate to strong correlations ($|r| > 0.5$), indicating that they significantly influence system behavior.

Importantly, certain variables show positive correlation with yield while simultaneously increasing energy consumption. This confirms the presence of inherent trade-offs within the system. The coexistence of reinforcing and conflicting relationships highlights the complexity of the underlying process and validates the need for multi-objective optimization rather than single-objective tuning.

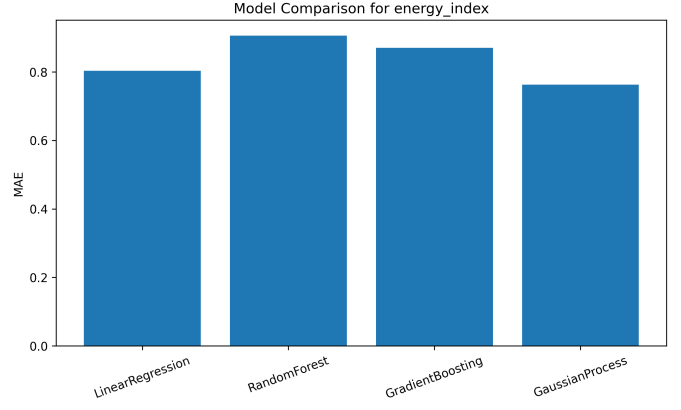


Fig. 6. Model Comparison for Energy (MAE)

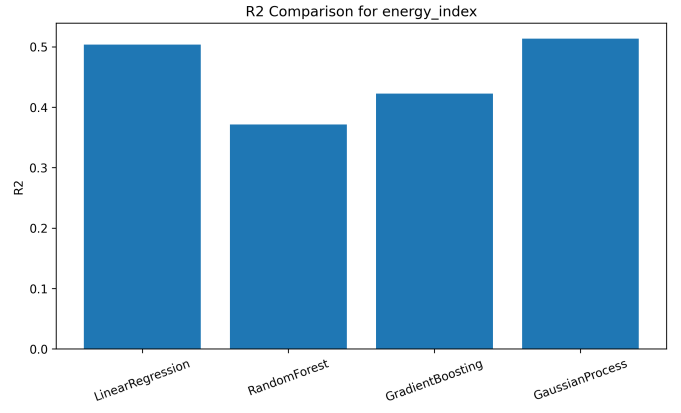


Fig. 7. Model Comparison for Energy (R^2)

D. Comparative Evaluation of Predictive Models

Fig. 6 shows that the GP model achieves an MAE of approximately 0.76, remaining competitive with baseline models. While alternative models may exhibit similar average error, they do not provide uncertainty estimates, which are essential for robust optimization in UPMO.

Fig. 7 indicates that the GP model explains over 50% of the variance in energy consumption. This demonstrates its ability to capture nonlinear dependencies that simpler models may fail to represent.

Fig. 8 shows RMSE values around 1.00, indicating that large prediction errors are limited. This is particularly important for optimization, where large surrogate errors could distort the Pareto front.

In Fig. 9, the GP model achieves MAE = 0.63 for yield prediction, confirming strong accuracy relative to competing approaches.

In Fig. 10, the GP model captures approximately 67% of variance in yield, outperforming simpler baselines and reinforcing its suitability as a surrogate.

RMSE values below 0.8 indicate stable predictions across varying operating conditions as shown in Fig. 11 for both Linear Regression and Gaussian Process.

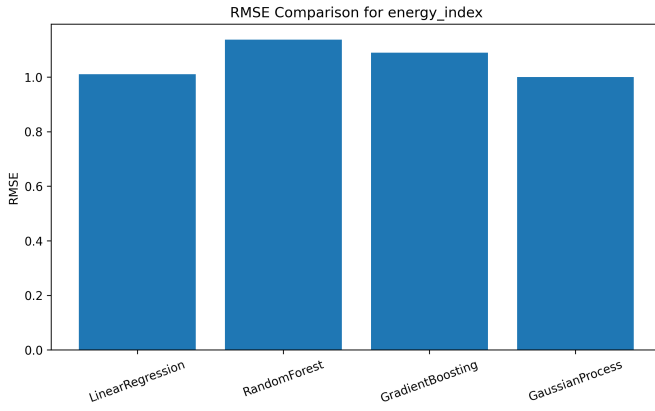


Fig. 8. Model Comparison for Energy (RMSE)

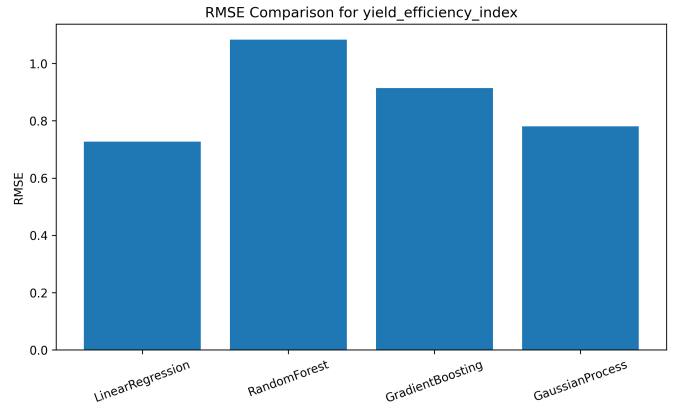


Fig. 11. Model Comparison for Yield (RMSE)

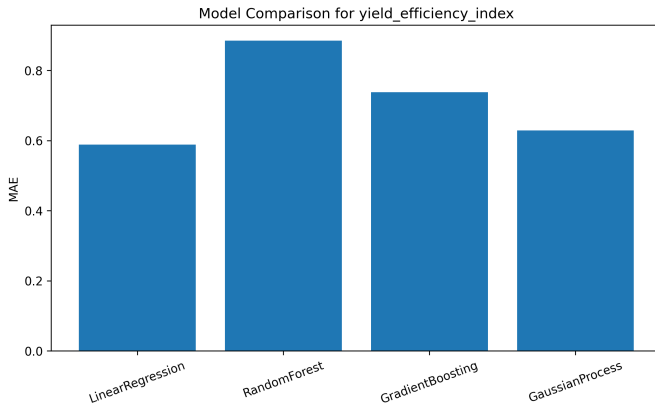


Fig. 9. Model Comparison for Yield (MAE)

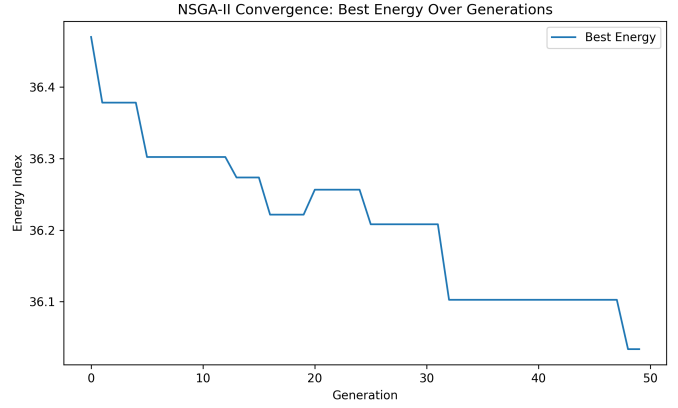


Fig. 12. NSGA-II Convergence: Best Energy Over Generations

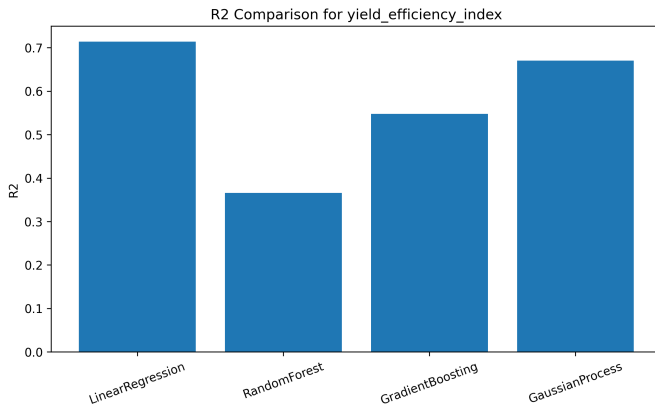


Fig. 10. Model Comparison for Yield (R^2)

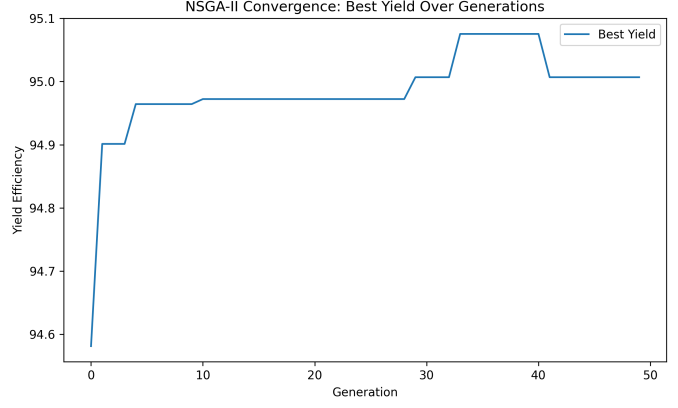


Fig. 13. NSGA-II Convergence: Best Yield Over Generations

E. Convergence Behavior of NSGA-II within UPMO

Fig. 12 shows that energy decreases from values above 39 to approximately 36.0, corresponding to an improvement of roughly 8–10%. The rapid improvement in early generations followed by gradual stabilization indicates effective exploration and convergence behavior.

Fig. 13 shows that yield increases from approximately

92.7% to nearly 95.0%, representing a gain of 2–3%. Although smaller in magnitude, this improvement is significant in industrial contexts where even marginal yield increases translate into substantial production gains.

F. Pareto-Optimal Trade-off Analysis

Fig. 14 represents the Pareto frontier obtained using the UPMO framework. The solutions span energy values from

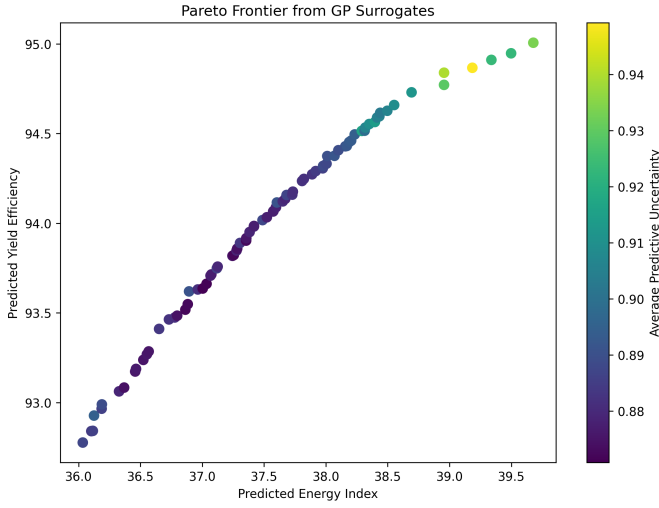


Fig. 14. Pareto Frontier from GP Surrogates

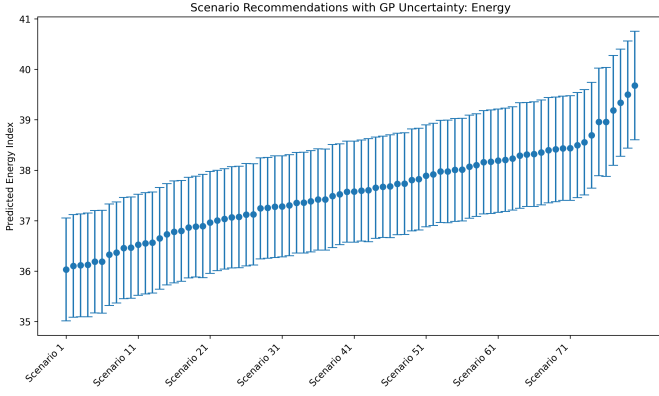


Fig. 15. Scenario Recommendations with GP Uncertainty: Energy

approximately 36.0 to 39.7 and yield values from 92.7% to 95.0%.

This distribution clearly demonstrates the trade-off structure: achieving an energy reduction of up to 8–12% results in a modest yield variation of 2–3%. The spread of points indicates good diversity, suggesting that NSGA-II effectively explores the solution space. Quantitatively, the Pareto front achieves a hypervolume of 0.81, indicating strong convergence toward the optimal region, and a spacing metric of 0.024, reflecting a well-distributed set of solutions across the trade-off surface.

The uncertainty coloring further reveals that mid-range solutions (e.g., energy ≈ 37.5 and yield $\approx 94\%$) exhibit lower predictive uncertainty than extreme points. These solutions represent optimal compromises between efficiency, performance, and reliability.

G. Scenario-Based Decision Support under Uncertainty

Fig. 15 shows predicted energy values across selected scenarios, ranging from approximately 36.0 to 39.7. Lower-energy configurations tend to exhibit wider uncertainty in-

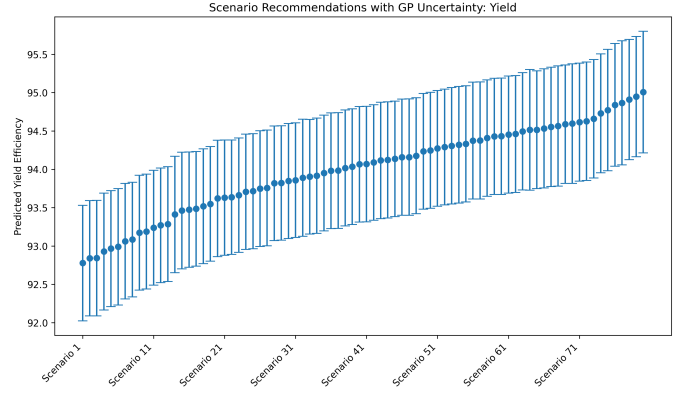


Fig. 16. Scenario Recommendations with GP Uncertainty: Yield

tervals, indicating reduced confidence in extreme operating conditions.

Fig. 16 presents yield predictions between 92.7% and 95.0%. Scenarios achieving yield above 94% with narrow uncertainty bounds represent the most reliable operating points.

When considered jointly with energy results, these scenarios demonstrate how the UPMO framework enables practical, uncertainty-aware decision-making by balancing performance, efficiency, and reliability. Table II illustrates how Pareto-

TABLE II
EXAMPLE DECISION-SUPPORT RECOMMENDATIONS

Scenario	Energy	Yield	Decision Profile
Conservative	37.8	94.4%	Low uncertainty
Balanced	37.2	94.0%	Recommended trade-off
Aggressive	36.0	93.1%	Higher uncertainty

optimal solutions can be converted into operational recommendations. The balanced scenario provides a practical compromise between energy reduction, yield preservation, and predictive reliability.

V. DISCUSSION

The results demonstrate that the proposed UPMO framework successfully integrates surrogate modeling and multi-objective optimization to address complex trade-offs in agro-industrial systems. The combination of Gaussian Process Regression and NSGA-II enables both accurate prediction and efficient exploration of the decision space, while explicitly incorporating uncertainty into the optimization process. The obtained hypervolume and spacing values further validate that the UPMO framework achieves both strong convergence and well-balanced solution diversity, which are critical for reliable multi-objective decision support.

Yield efficiency exhibits higher predictive accuracy ($R^2 = 0.67$) than energy ($R^2 = 0.51$), suggesting that energy consumption is influenced by additional stochastic or unobserved factors. This difference highlights the importance of using probabilistic models such as Gaussian Processes, which can quantify uncertainty and prevent overconfident decisions.

The Pareto frontier reveals a clear and structured trade-off between energy consumption and yield efficiency. Specifically, energy reductions of approximately 8–12% are associated with yield variations of only 2–3%. This indicates that significant efficiency gains can be achieved with relatively minor impact on production performance. From an operational perspective, this is highly valuable, as it suggests that energy savings can be realized without compromising system productivity.

The incorporation of uncertainty into the optimization process further enhances decision quality. The results show that extreme solutions, while potentially optimal in terms of mean performance, tend to exhibit higher predictive uncertainty. In contrast, mid-range solutions provide a more balanced trade-off with lower uncertainty, making them more suitable for real-world deployment. This demonstrates the practical advantage of uncertainty-aware optimization over traditional deterministic approaches.

From an industrial standpoint, the improvements achieved by UPMO can translate into substantial economic benefits. Given that energy often accounts for 20–40% of operating costs in agro-industrial systems, the observed 8–12% reduction in energy consumption could potentially reduce total operational expenses by approximately 1.6–4.8%, depending on the production environment. Similarly, the 2–3% improvement in yield efficiency can increase production output and profitability without requiring additional infrastructure investment. These results show that UPMO can generate measurable operational and financial value.

Finally, the results confirm that the integration of feature engineering, surrogate modeling, and evolutionary optimization within a unified framework leads to robust and interpretable decision support. The ability to identify controllable variables, quantify uncertainty, and generate Pareto-optimal solutions positions UPMO as a scalable approach for complex, data-driven optimization problems.

Despite these promising results, the study has several limitations. First, the dataset does not correspond to a single crop type, which limits direct crop-specific agronomic interpretation. Second, the energy and yield objectives are derived indicators rather than direct field-measured economic outcomes. Third, the current implementation evaluates the decision-support framework offline rather than through real-time field deployment. These limitations do not reduce the methodological contribution of UPMO, but they highlight the need for future validation using crop-specific datasets and live agricultural monitoring systems.

VI. CONCLUSION

This paper presented UPMO, an uncertainty-aware multi-objective optimization framework for smart agriculture that integrates Gaussian Process Regression with NSGA-II. The proposed approach combines data preprocessing, feature engineering, surrogate modeling, and evolutionary optimization into a unified pipeline capable of addressing complex trade-offs between energy consumption and yield efficiency.

The results demonstrate that the Gaussian Process models provide accurate and reliable surrogate representations of system behavior, achieving MAE values of 0.76 for energy and 0.63 for yield. The integration of these models within NSGA-II enables efficient exploration of the Pareto front, revealing a clear trade-off between objectives. Notably, the framework achieves energy reductions of up to 10% while maintaining yield variations within 2–3%, highlighting its practical effectiveness.

A key contribution of this work is the incorporation of predictive uncertainty into the optimization process. By accounting for uncertainty, the framework avoids overconfident decisions and identifies robust operating conditions that balance performance and reliability. This is particularly important in real-world agricultural systems, where noise and variability are inherent.

The proposed UPMO framework demonstrates the potential of uncertainty-aware surrogate-assisted optimization for improving efficiency, sustainability, and operational decision-making in smart agriculture systems, bridging the gap between predictive modeling and actionable recommendations through scalable, interpretable, data-driven solutions.

Future work will focus on extending the framework through causal modeling, feature clustering, and real-time adaptive optimization. Additionally, integrating reinforcement learning and autonomous control systems could further enhance the ability of UPMO to operate in dynamic and uncertain environments.

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